Hartshorne Chapter 2 Section 3: Properties of Schemes

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Table of Content:

The main goal for this section is understanding the topological / algebraic properties on schemes.

1 Basic Properties, and their Relations

Definition: Connected Scheme (Topological)

A scheme is **connected** if it as topological space is connected.

Definition: Irreducible Scheme (Topological)

A scheme is **irreducible** if it as topological space is connected.

Definition: Reduced Scheme (Algebraic)

A scheme is **reduced** if every open subset $U \subseteq X$ has $\mathcal{O}_X(U)$ being reduced.

Definition: Integral Scheme (Algebraic)

A scheme is **integral** if every open subset $U \subseteq X$ has $\mathcal{O}_X(U)$ being an integral domain.

As an examples / exercise, here are some properties of affine schemes:

Proposition

Given $X = \operatorname{Spec}(A)$ an affine scheme, then:
1. X is connected iff $A \ncong B \times C$ as commutative rings.
2. X is irreducible iff $\operatorname{Nil}(A)$ is prime
3. X is reduced iff $\operatorname{Nil}(A) = 0$
4. X is integral iff A is an integral domain.

Proof:

1. It's easier to prove when assuming A is already reduced, and also X is disconnected iff $A \cong B \times C$ as commutative rings.

   If X is disconnected (), then consider the two radicals $I, J \subseteq A$ that satisfies $X = V(I) \sqcup V(J)$. Since $V(I) \cap V(J) = V(I + J) = \emptyset$, one must have $I + J = A$; on the other hand, $V(I) \sqcup V(J) = V(I \cap J) = X = \operatorname{Spec}(A)$ (while I, J are radicals, implying $I \cap J$ is radical) shows that $I \cap J = \mathrm{Nil}(A) = 0$ (by assumption). So, one has $A \cong I \oplus J$ as A-module, in particular each I, J is itself a ring (since $J \cong A/I$).

Then, for non-reduced case, if $A \twoheadrightarrow A/\mathrm{Nil}(A) \cong \overline{B} \times \overline{C}$, let B, C be the preimage of $\overline{B}, \overline{C}$ respectively, we ought to show $B \times C \twoheadrightarrow A$ (then since the ideal in $B \times C$ are of the form $I_B \times I_C$, one has $A \cong B/I_B \times C/I_C$, where $I_B \times I_C$ is the kernel). However, it's simple by $(b, c) \mapsto b + c$

Suppose $A \cong B \times C$, then $B \hookrightarrow B \times C$ and $C \hookrightarrow B \times C$ are ideals, satisfying $B + C = A$ and $B \cap C = 0$, showing $V(B) \cap V(C) = V(B + C) = V(A) = \emptyset$, and $V(B) \sqcup V(C) = V(B \cap C) = V(0) = \mathrm{Spec}(A)$. So, $\mathrm{Spec}(A)$ is disconnected.

2. Since one has $\mathrm{Spec}(A) = V(\mathrm{Nil}(A))$, then one has $\mathrm{Spec}(A) \cong \mathrm{Spec}(A/\mathrm{Nil}(A))$ as topological space. Which, if $\mathrm{Nil}(A)$ is prime, $A/\mathrm{Nil}(A)$ is an integral domain, hence $\mathrm{Spec}(A)$ is irreducible; conversely, suppose $\mathrm{Spec}(A) \cong \mathrm{Spec}(A/\mathrm{Nil}(A))$ is irreducible, but $\mathrm{Nil}(A)$ not integral domain, there exists $i, j \in A \setminus \mathrm{Nil}(A)$, such that $\bar{i} \cdot \bar{j} = 0$. As a result, one has $\mathrm{Spec}(A/\mathrm{Nil}(A)) = V((0)) = V\big((\bar{i})(\bar{j})\big) = V\big((\bar{i})\big) \cup V\big((\bar{j})\big)$, while each $V\big((\bar{i})\big), V\big((\bar{j})\big)$ are proper, contradicting the irreducibility. Hence, $\mathrm{Nil}(A)$ must be prime.

3. For reduced property, we'll verify that they're local properties first:
   - First, recall that $\mathrm{Nil}(A_P) = (\mathrm{Nil}(A)_P)$ (by the fact that extension of ideals under localization preserves intersection, and any other prime ideals not containing P extends to unit ideal).

     Hence, one has $\mathrm{Nil}(A) = 0$ (as A-modules) $\iff \mathrm{Nil}(A_P) = \mathrm{Nil}(A)_P = 0$ (as A_P-module) for all prime ideal $P \subseteq A$ (this is the module property).

   Now, since every open subset $U \subseteq X$ has $\mathcal{O}_X(U) = \left\{ s : U \to \bigsqcup_{P \in U} A_P \right\}$, since A is reduced $\iff A_P$ being reduced, it enforces $\mathcal{O}_X(U)$ to be reduced also (since to be nilpotent it's cooredinate-wise nilpotent, which is 0).

   Conversely, suppose A is not reduced, then one of A_P has some $a \in A$ such that $\frac{a}{1} \neq 0$ but $\left(\frac{a}{1}\right)^n = 0$ in A_P, or $a^n u = 0$ for some $u \in A \setminus P$. Then, take the open subset $D(u)$, one has the function $s : D(u) \to \bigsqcup_{Q \in D(u)} A_Q$ by $s(Q) = \frac{a}{1}$ satisfies $\left(\frac{a}{1}\right)^n = \frac{a^n}{1} = 0$ (since $a^n u = 0$, with $u \notin Q$ for all $Q \in D(u)$). Then, $s \neq 0$ (since $s(P) = \frac{a}{1} \neq 0 \in A_P$), while $s^n = 0$. So, $\mathcal{O}_X(D(u)$ is not reduced.

4. If X is integral, it's obvious that $\mathcal{O}_X(X) = A$ is integral domain.

   The converse will be proven in the next proposition (for general scheme) (where a scheme is integral $\iff$ it's reduced and irreducible. Here, reduced and irreducibility both come from the assumption that A is integral domain).

(**Remark:** The general case for non-reduced version of 1 is not solved.) $\square$

Now, here's the general statement on the scheme:

Proposition

A scheme X is integral iff it is both reduced and irreducible.

Proof: $\Longrightarrow$:

Suppose X is integral, it's clear that it's reduced (as all integral domain is reduced).

For irreducibility, suppose the contrary that it's reducible. Hence, there exists two nonempty open subsets $U, V \subseteq X$, where $U \cap V = \emptyset$. As a result, take any $s_u \in \mathcal{O}_X(U)$ and $s_v \in \mathcal{O}_X(V)$, since their domain has empty intersection, they can always be glued together. So, take $(s_u, 0_v)$ as the gluing of nonzero s_u with $0_v \in \mathcal{O}_X(V)$, $(0_u, s_v)$ as the gluing of nonzero s_v with $0_u \in \mathcal{O}_X(U)$. The product of the two is 0 (as 0_u evaluates to 0 on U, and similarly for 0_v), while each individual of them is not, reaching a contradiction that $\mathcal{O}_X(U \sqcup V)$ is an integral domain.

(Even stronger, one can say $\mathcal{O}_X(U \sqcup V) \cong \mathcal{O}_X(U) \times \mathcal{O}_X(V)$ if $U \cap V = \emptyset$, as the two functions share no domain, so has no affect on each other).

$\Longleftarrow$:

Conversely, suppose X is reduced and irreducible. Let $U \subseteq X$ be any open subset, and suppose $f, g \in \mathcal{O}(U)$ satisfies $fg = 0$. Define the set $Z_f := \{x \in U \mid f_x \in m_x\}$ (where $f_x \in \mathcal{O}_{X,x}$ and m_x is the maximal ideal of $\mathcal{O}_{X,x}$), and similarly $Z_g := \{x \in U \mid g_x \in m_x\}$. Note that since $f_x g_x = 0 \in m_x$ for all $x \in U$, each $x \in U$ has $f_x \in m_x$ or $g_x \in m_x$, showing $Z_f \cup Z_g = U$.

On the other hand, they're actually closed subsets: By intersecting with affine covers, say U is affine (then $U = \mathrm{Spec}(\mathcal{O}_X(U))$), then the complement $X_f := U \setminus Z_f$ are all prime ideals $P \in U$, such that f_P is a unit (in particular, one has $f \notin P$, as $\mathcal{O}_{X,P}$ is the localization of $\mathcal{O}_X(U)$ at P, then f_P is a unit $\Longrightarrow f$ can't be contained in P). So, working it out set wise, we have $X_f = D(f)$ in fact, which is open in U. So, Z_f is closed in U. And, since this is true for arbitrary affine cover, Z_f must be closed in X itself.

So, we have Z_f, Z_g two closed subset such that $Z_f \cup Z_g = U$. With X irreducible, any of its open subset must also be irreducible, showing $Z_f = U$ or $Z_g = U$.

Suppose $Z_f = U$, then this implies $X_f = D(f) = \emptyset$ when restricting it to U's intersection with any affine cover, as a result f must be nilpotent in all of these affine covers, hence nilpotent in U. But, using the reduced assumption, this implies $f = 0$. Which, it proves the integrality. $\square$

Intuition wise, we can see it relies heavily on affine schemes. If possible, in the future we'll practice more on how restricting to affine scheme works.

2 Noetherian Property

Similar to how in classical sense we worked with $k[x_1, ..., x_n]$ (a Noetherian ring), we wish to work with some similar ideas in scheme theory also.

Definition: Locally Noetherian, Noetherian Scheme

A scheme X is **locally Noetherian**, if there exists an open cover $\{U_i\}$ of X, such that each $U_i = \operatorname{Spec}(A_i)$ (affine), and each A_i is a Noetherian ring.

Similarly, X is **Noetherian**, if it's locally Noetherian and quasi-compact. As an equivalent definition, X is Noetherian, if it can be covered by finitely many affine open subsets, with each affine subset corresponds to a Noetherian ring.

Remark

If X is a Noetherian scheme, then it's clear it's Noetherian as topological space (since it's finite union of $\operatorname{Spec}(A_i)$, which A_i is Noetherian ring guarantees $\operatorname{Spec}(A_i)$ is a Noetherian topological space; so, finite union is still Noetherian topological space).

Yet, the converse is false:

Consider the ring $A = k[x_1, x_2, ...]/(x_1^2, x_2^2, ...)$. Notice that all $\overline{x_i} \in A$ are nilpotent, hence $(\overline{x_1}, \overline{x_2}, ...) \subseteq \operatorname{Nil}(A)$; on the other hand, it's also a maximal ideal (as the quotient $k[x_1, x_2, ...] \twoheadrightarrow k$ by sending all $x_i \mapsto 0$ contains the ideal $(x_1^2, x_2^2, ...)$, hence factors through A; however, in here any constant can't be sent to 0, while all $\overline{x_i} \mapsto 0$, so the kernel through A must be $(\overline{x_1}, \overline{x_2}, ...)$, showing that $A/(\overline{x_1}, \overline{x_2}, ...) \cong k$, hence it's also maximal). As a result, $\operatorname{Spec}(A) = \{*\}$ (since it must only be the maximal ideal), which is trivially Noetherian.

On the other hand, the ring A can never be Noetherian, since the chain $(\overline{x_1}) \subset (\overline{x_1}, \overline{x_2}) \subset ...$ pulls back to $(x_1) \subset (x_1, x_2) \subset ...$ in $k[x_1, x_2, ...]$ (which is a strict chain), so it must also be a strict chain in A, showing it's not Noetherian as ring.

Finally, $\operatorname{Spec}(A)$ is not Noetherian as schemes, since the only open affine subset is itself, and the ring is A (which is not Noetherian).

This definition is good in an algebraic sense, as one can always find an affine neighborhood associated to a Noetherian ring. Yet, it'll be a lot more convenient if one can choose any affine open neighborhood to be associated to Noetherian rings. This is indeed the case in general:

Proposition

A scheme X is locally Noetherian iff every open affine subset $U \subseteq X$, $U \cong \operatorname{Spec}(A)$ has A being a Noetherian ring.

In particular, if $X = \operatorname{Spec}(A)$, then it's a Noetherian scheme iff A is Noetherian.

Proof: $\Longleftarrow$:

Suppose every open affine subset $U \cong \operatorname{Spec}(A)$ for some Noetherian ring A, it's clear X is a locally Noetherian scheme by definition.

$\Longrightarrow$:

Suppose X is locally Noetherian, and given any open affine subset $U \cong \operatorname{Spec}(A)$ (i.e. the restriction of schemes has $(U, \mathcal{O}_U) \cong (\operatorname{Spec}(A), \mathcal{O}_A)$ as locally ringed space). Choose $\{U_i\}$ as the affine cover of X so that each $U_i \cong \operatorname{Spec}(B_i)$ for some Noetherian B_i, and consider the intersection $U \cap U_i \subseteq \operatorname{Spec}(B_i)$. Since within $\operatorname{Spec}(B_i)$ this is open, by quasi-compactness of prime spectrum, there exists finitely many $f_{i,1}, ..., f_{i,j_i} \in B_i$, so that $U \cap U_i = D(f_{i,1}) \cup ... \cup D(f_{i,j_i})$, and each $D(f_{i,l})$ has $\mathcal{O}_A(D(f_{i,l})) \cong (B_i)_{f_{i,l}} \cong B_i[x]/(xf_{i,l} - 1)$, which by the Noetherianness of B_i, this ring is also Noetherian.

As a result, we swap $U \cong \operatorname{Spec}(A)$ with all of the following:

$$U = \bigcup_i U \cap U_i = \bigcup_i \left(\bigcup_{l=1}^{j_i} D(f_{i,l}) \right) \tag{2.1}$$

Where, each $D(f_{i,l})$ corresponds to a Noetherian ring (and by the isomorphism of schemes, one has $\mathcal{O}_X(D(f_{i,l})) \cong (B_i)_{f_{i,l}}$). Hence, the statement also reduces to: If $X = \operatorname{Spec}(A)$ is an affine scheme covered by affine opens that're Noetherian rings' spectra, then A is Noetherian.

Suppose $U \subseteq \operatorname{Spec}(A) = X$ is an affine open subset where $U \cong \operatorname{Spec}(B)$ (with B Noetherian ring), this inclusion $U \hookrightarrow X$ induces a ring homomorphism $A \to B$. Now, by openness of U, by the fundamental open sets, let $f \in A$ satisfies $D(f) \subseteq U$; let $\overline{f} \in B$ be the image of f under the ring homomorphism, then the fundamental open's property shows that $\iota^{-1}(D(f)) = D(\overline{f})$, in particular one has $\mathcal{O}_B(D(\overline{f})) \cong \mathcal{O}_A(D(f))$, so $B_{\overline{f}} \cong A_f$, showing A_f is Noetherian.

This again swaps X with a collection of $f \in A$, such that $D(f)$ associates to A_f a Noetherian ring. Using the quasi-compactness of $X = \operatorname{Spec}(A)$, there exists $f_1, ..., f_n \in A$, such that $\bigcup_{i=1}^n D(f_i) = X$, and each A_{f_i} is Noetherian. As a result, $(f_1, ..., f_n) = A$. Now, let's do a lemma:

Lemma

Let $I \subseteq A$ be an ideal, and $\varphi_i :\to A_{f_i}$ be the localization map for each $i \in \{1, ..., n\}$. Then:

$$I = \bigcap_{i=1}^n I^{ec} = \bigcap_{i=1}^n \varphi_i^{-1}\left(\varphi_i(I) A_{f_i}\right) \tag{2.2}$$

Proof: Let's recall that each $\varphi_i^{-1}\left(\varphi_i(I)A_{f_i}\right) = \left\{r \in A \mid \exists k \in \mathbb{N}, \quad r f_i^k \in I\right\}$ (since the extension then contraction is all the element, such that multiplied by some element in the multiplicative set it lies in I). So, this shows the inclusion $I \subseteq \bigcap_{i=1}^n \varphi_i^{-1}\left(\varphi_i(I)A_{f_i}\right)$.

Now, suppose $r \in \bigcap_{i=1}^n \varphi_i^{-1}\left(\varphi_i(I)A_{f_i}\right)$, then there exists $k_1, ..., k_n \in \mathbb{N}$ and $a_1, ..., a_n \in I$, such that $\varphi_i(r) = \frac{a_i}{f_i^{k_i}} \in A_{f_i}$. In particular, by multiplying suitable $f_i^{l_i}$ to both the numerator and denominator, WLOG one can assume all $k_i = k$ for some fixed k. So, we have $\varphi_i(r) = \frac{r}{1} = \frac{a_i}{f_i^k}$, showing there exists $m_1, ..., m_n \in \mathbb{N}$, such that $\left(f_i^k r - a_i\right) f_i^{m_i} = 0$. Now, by multiplying new f_i's wouldn't affect the equality, hence again WLOG each $m_i = m$ for some fixed m.

Herefore, we get $\left(f_i^k r - a_i\right) f_i^m = 0$ for all index i, showing $f_i^{k+m} r \in I$. Now, since $(f_1, ..., f_n) \in A$, notice that $\left(f_1^{k+m}, ..., f_n^{k+m}\right) = A$ (since its radical contains $f_1, ..., f_n$, hence equals to A). Then, for suitable $b_1, ..., b_n \in A$, one has the following:

$$\sum_{i=1}^n b_i f_i^{k+m} = 1, \quad \sum_{i=1}^n b_i f_i^{k+m} r = r \in I \tag{2.3}$$

This finishes the proof. □

As a result, for any increasing chain of ideals $I_1 \subset I_2 \subset ...$, its image $\varphi_i(I_1)A_{f_i} \subset \varphi_i(i_2)A_{f_i} \subset ...$ in A_{f_i} stabilizes (by a.c.c. on A_{f_i}), hence $\varphi_i^{-1}\left(\varphi_i(I_1)A_{f_i}\right) \subset \varphi_i^{-1}\left(\varphi_i(I_2)A_{f_i}\right) \subset ...$ stabilizes in A.

Then, the lemma states that each I_i is the intersection of the ith ideal of a stabilizing chain, hence for large enough N the I_i's chain stabilizes. This shows that A is Noetherian. □

For the spcial case where $X = \operatorname{Spec}(A)$ is a Noetherian Scheme, locally Noetherianness enforces all rings of affine open subsets to be Noetherian, in particular $\mathcal{O}_X(X) = A$ is Noetherian.

This demonstrates another important mindset in Algebraic Geometry: Sometimes we're looking at information beyond the space, so being able to transfer between algebraic properties and geometric properties will be crucial.

3 Morphisms

Now, we'll start talking about some basic morphisms. Recall that for classical sense, the finite type / finite ring homomorphisms can let us reduce the questions to some quotient of polynomial rings / realizing it as finitely-generated module over the base ring. Here it's similar in scheme theory:

Definition: Locally Finite Type / Finite Type Morphism

Given a morphism of schemes $f : X \to Y$, it's called **locally of finite type** if there exists an affine cover $\{V_i\}$ of Y (denote each $V_i = \operatorname{Spec}(A_i)$), such that each $f^{-1}(V_i)$ has an affine cover $\{U_{ij}\}$ (denote each $U_{ij} = \operatorname{Spec}(A_{ij})$), which the restricted morphism $f : U_{ij} \to V_i$ induces a ring homomorphism $f^\# : A_i \to A_{ij}$ that realizes A_{ij} as a finitely-generated A_i-algebra.

Similarly, f is called **finite type** if the covering $f^{-1}(V_i) = \bigcup_j U_{ij}$ can be chosen using finite number of j.

Definition: Finite Morphism

Given a morphism of schemes $f : X \to Y$, it's called **finite** if there exists a affine cover $\{V_i\}$ of Y (denote each $V_i = \operatorname{Spec}(A_i)$), such that each $f^{-1}(V_i)$ is affine (denote $F^{-1}(V_i) = \operatorname{Spec}(B_i)$), where the restricted morphism $f : f^{-1}(V_i) \to V_i$ induces a ring homomorphism $f^\# : A_i \to B_i$, realizing B_i as a finitely-generated A_i-module.

Again, similar to the previous definition of Locally Noetherian, it in fact enforces all affine open subsets to have the same properties (the proofs are highly nontrivial though...)

Now, let's dive into some properties on open / closed subschemes:

Definition: Open Subscheme, Open Immersion

Let X be a scheme, a scheme U is an **open subscheme** of X, if there exists an open subset $U' \subseteq X$, such that as schemes one has $(U, \mathcal{O}_U) \cong (U', \mathcal{O}_X|_{U'})$.

An **open immersion** is a morphism $f : X \to Y$, such that it can be recognized as the inclusion $\iota : U \hookrightarrow X$ of open subscheme U (or, an isomorphism of open subschemes).

Definition: Closed Immersion, Closed Subscheme

A morphism of schemes $f : X \to Y$ is a **closed immersion** if as topological space, X is homeomorphic to $f(X)$ that is a closed subset of Y, while the induced sheaf morphism $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective.

A **closed subscheme** of Y is a scheme X together with a closed immersion $f : X \to Y$. This is also classified up to isomorphism.

As of examples, we have the following:

Example: Ring Projection

Given any projection of rings $A \twoheadrightarrow A/I$, it induces a homeomorphism $f : \operatorname{Spec}(A/I) \xrightarrow{\sim} V(I) \hookrightarrow \operatorname{Spec}(A)$, which is a closed map in topological sense; the reason why it's surjective on the morphism of sheaves, is because for any prime ideal $P \subseteq A/I$, with prime ideal $P^c \subseteq A$, one has the following commutative diagram:

$$\begin{tikzcd} A \arrow[r, two heads] \arrow[d] & A/I \arrow[d] \\ A_{P^c} \arrow[r, two heads] & (A/I)_P \cong A_{P^c}/I_{P^c} \end{tikzcd}$$

Where, the property that localization and quotient commutes (as long as the ideal does't touch the multiplicatively closed set) shows the bottom map is surjective.

Hence, the morphism $\mathcal{O}_A \to f_* \mathcal{O}_{A/I}$ is surjective as sheaves (since it's surjective on stalks, while stalks preserve exactness).

We'd also like to talk about how to „induce" a scheme structure on a closed subset of a scheme. For open subset it's clear that the choice must be the restriction to subspace topology + the restriction of sheaf, but since there's no notion of sheaf over closed subsets, we do need a bit more work.

Example: Reduced Induced Closed Subscheme of Affine Scheme

Given $X = \operatorname{Spec}(A)$ an affine scheme, and $Y \subseteq X$ be a closed subset. Then, $Y = V(I)$ for some ideal I, which one can take $Y = V\left(\sqrt{I}\right)$ (the largest ideal that doesn't change Y). Which, using the homeomorphism $V\left(\sqrt{I}\right) \cong A/\sqrt{I}$, this induces a reduced scheme structure on Y using $\sqrt{I}$.

For general schemes, we require the gluing of schemes as description.

Example: Structure on General Schemes

Let X be arbitrary scheme, and $Y \subseteq X$ a closed subset. Let $\{U_i\}$ be an affine cover of X (with $U_i = \operatorname{Spec}(A_i)$), and consider $Y_i = Y \cap U_i$ (which is closed in U_i), which has an induced reduced structure given for the affines (this structure may be dependent on U_i).

Now, to glue the schemes together, we need to argue that any $Y_i \cap Y_j \subseteq Y_i$ is isomorphic to $Y_i \cap Y_j \subseteq Y_j$ (since they may induce different structures). Since $Y_i \cap Y_j = Y \cap \left(U_i \cap U_j\right) = Y_i \cap U_j$, one needs to prove the induced structure is independent of the order of restriction (i.e. one can restrict to one then the other, or vice versa). This suffices to prove the case for the fundamental open subsets.

The full statement is: Given $U = \operatorname{Spec}(A)$, if $f \in A$ and $V = D(f) = \operatorname{Spec}\left(A_f\right)$, then the reduced induced structure on $Y \cap U$ restricts to $Y \cap V$ agrees with $Y \cap V$'s reduced induced structure. However, given any ideal $I \subseteq A$ that's a radical, one has $I = \bigcap_{P \in V(I)} P$ in A, when extending it becomes $I^e = \bigcap_{P \in V(I)} P^e$ in A_f, which P^e is prime $\iff$ its intersection with the set of f^n's is empty, or $P \in D(f)$. So, going from $Y \cap U$ (as $V(I)$) to $Y \cap V$ (as $V(I^e)$) under extension, is the same as the structure on $Y \cap V$ (as $V(I) \cap D(f)$).

Finally, the gluing condition of sheaves ensure the existence of a structure after gluing all affines together.

This unfortunately is not mentioned in class, which is a bit annoying (since it's still important in our sense).

4 Fibre Product

This will be the most complicated part in this section, as the construction is highly nontrivial. The goal is to get the categorical fibre prodct in the category of schemes. Similar to what we've been doing, it's the matter of reducing down to the fibre products between affine schemes. For this, let's focus on one tool first:

Theorem: Affinization

Given the functor $\mathrm{Spec}(\_) : \mathsf{CRing}^{\mathrm{op}} \to \mathsf{Sch}$, it has a left adjoint, named „Affinization functor" $\mathrm{Aff}_{(\_)} : \mathsf{Sch} \to \mathsf{CRing}^{\mathrm{op}}$, that's based on the universal property:

Given any scheme X, its affinization Aff_X is an affine scheme with a morphism of schemes $f : X \to \mathrm{Aff}_X$, such that for any affine scheme Y and morphism of schemes $g : X \to Y$, there exists a nique morphism of affine schemes $\overline{g} : \mathrm{Aff}_X \to Y$, such that $g = \overline{g} \circ f$.

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "\forall g"'] & & \mathrm{Aff}_X \arrow[dl, dotted, "\exists! \overline{g}"] \\ & Y & \end{tikzcd}$$

This provides a natural isomorphism as follow (where $\mathrm{Aff}_X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(A)$):

$$\mathrm{Hom}_{\mathsf{CRing}^{\mathrm{op}}}(B, A) \cong \mathrm{Hom}_{\mathsf{AffSch}}(\mathrm{Aff}_X, Y) \cong \mathrm{Hom}_{\mathsf{Sch}}(X, Y) \tag{4.1}$$

Proof: For any scheme X, define the ring $A_X := \mathcal{O}_X(X)$ (i.e. the ring associated to the global section, A_X is often denoted as $\Gamma(X < \mathcal{O}_X)$, called the „global section functor"). And, define $\mathrm{Aff}_X := \mathrm{Sp}(A_x)$.

Which, given any affine scheme $Y = \mathrm{Spec}(B)$, and morphism of schemes $f : X \to Y$, it associates with the morphism of sheaves $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$, which $h = f^\#_Y : \mathcal{O}_Y(Y) \to \mathcal{O}_X(f^{-1}(Y)) = \mathcal{O}_X(X)$ is a ring homomorphism $h : B \to A_x$, hence reverses to a unique morphism of schemes on the spectrum level $h^* : \mathrm{Spec}(A_X) = \mathrm{Aff}_X \to \mathrm{Spec}(B) = Y$. So, if such map exists, it must be unique.

Now, we need to show: Given any ring homomorphism $\alpha : B = \mathcal{O}_Y(Y) \to A_X = \mathcal{O}_X(X)$, it defines a morphism of schemes $\alpha^* : X \to Y$. For this, we need gluin of schemes again.

Suppose $\{U_i\}$ collects all affine open subsets of X, each $U_i = \mathrm{Spec}(A_i)$, then the inclusion $\iota_i : U \hookrightarrow X$ together with $f : X \to Y$ composes to $f \circ \iota_i : U_i \to Y$, a morphism between affine schemes, hence it reverses to a ring homomorphism $\alpha_i^* : B = \mathcal{O}_Y(Y) \to A_i = \mathcal{O}_X(U)$. Also, the way we define the morphism enforces this to be coming from the composition of $f^\#_Y : B = \mathcal{O}_Y(Y) \to A = \mathcal{O}_X(X)$ and $\iota_i^* : A = \mathcal{O}_X(X) \to A_i = \mathcal{O}_X(U)$:

$$\begin{tikzcd} U_i \arrow[rr, hook, "\iota_i"] \arrow[dr, "f \circ \iota_i"'] & & X \arrow[dl, "f"] \\ & Y & \end{tikzcd} \qquad \begin{tikzcd} A_i & & A \arrow[ll, "\iota_i^*"'] \\ & B \arrow[ul, "\alpha_i^*"] \arrow[ur, "f^\#_Y"'] & \end{tikzcd}$$

Now, consider $U_i \cap U_j$. As a remark, it may not be affine, but it has affine neighborhoods in each U_i, U_j, so WLOG assume we're working with the affine neighborhood $W \subseteq U_i \cap U_j$. Which, we claim that the structure on W is independent of U_i, U_j, and this is because for the two inclusions $W \hookrightarrow U_i \hookrightarrow X$ and $W \hookrightarrow U_j \hookrightarrow X$, the restriction of sheaves forces $\mathcal{O}_W(W) = \mathcal{O}_X(W)$, independent of which U_i it passes through.

In particular, this forces the ring homomorphism $\alpha_W^* : A = \mathcal{O}_Y(Y) \to \mathcal{O}_W(W)$ to be compatible in the following sense (where the top is morphism of schemes, the bottom is ring homomorphisms):

$$
\begin{tikzcd}
 & U_i \arrow[dr, "f\circ\iota_i"] \arrow[d, hook, "\iota_i"] & \\
W \arrow[ur, hook, "\iota_{W,i}"] \arrow[r, hook, "\iota_W"] \arrow[dr, hook, "\iota_{W,j}"'] & X \arrow[r, "f"] & Y \\
 & U_j \arrow[u, hook, "\iota_j"'] \arrow[ur, "f\circ\iota_j"'] &
\end{tikzcd}
$$

$$
\begin{tikzcd}
 & A_i \arrow[dl, "\iota^*_{W,i}"'] & \\
\mathcal{O}_W(W) & A \arrow[l, "\iota^*_W"'] \arrow[u, "\iota^*_i"] \arrow[d, "\iota^*_j"'] & B \arrow[l, "f^\#_Y"'] \arrow[ul, "\alpha^*_i"'] \arrow[dl, "\alpha^*_j"] \\
 & A_j \arrow[ul, "\iota^*_{W,j}"] &
\end{tikzcd}
$$

So, it's a really large commutative diagram chase between CRing and Sch.

Which, the morphism $W \to Y$ induces a clear ring homomorphism $B \to O_W(W)$ that's compatible with the restriction (and intersection), showing the morphisms could glue together (in a unique way). Globally, this is the morphism $\alpha : X \to Y$ (and the property forces it to be unique).

□

With this, we can define the fibre product over affine schemes:

Lemma: Fibre Product of Affine Schemes

Given three affine affine schemes X, Y, Z (corresponds to ring C, B, A respectively), and morphisms of schemes $f : X \to Z$, $g : Y \to Z$ (corresponds to ring homomorphisms $f^* : A \to C$, $g^* : A \to B$ respectively), then the fibre product in Sch exists.

Proof: Note that we seek for something stronger: Instead of fibre product over AffSch, it's fibre product over Sch for any morphisms between affine schemes (so, the universal property must work for any schemes, not just affine schemes). Nonetheless, we can start with just affine schemes:

Since we have the ring homomorphisms going out from A, it's suitable to construct the fibre coproduct in CRing, namely the tensor product:

$$
\begin{tikzcd}
A \arrow[r, "f^*"] \arrow[d, "g^*"'] & C \arrow[d, "(g^*)'"] \arrow[ddr, bend left, "d^*_c"] & \\
B \arrow[r, "(f^*)'"'] \arrow[drr, bend right, "d^*_b"'] & B \otimes_A C \arrow[dr, dotted, "\exists ! h^*"] & \\
 & & D
\end{tikzcd}
$$

Which, we define $X \times_Z Y := \operatorname{Spec}(B \otimes_A C)$, and $f' : X \times_Z Y \to Y$, $g' : X \times_Z Y \to X$ be induced by $(f^*)', (g^*)'$ respectively. Then, the commutativity of the ring homomorphism guarantees the following diagram:

$$\begin{tikzcd}
Z & X \arrow[l, "f"'] \\
Y \arrow[u, "g"] & X \times_Z Y \arrow[l, "f'"] \arrow[u, "g'"']
\end{tikzcd}$$

And, over AffSch, it satisfies the universal propety, since for any affine schemes W (say corresponds to ring D), together with morphisms of schemes $d_B : W \to Y$ and $d_C : W \to X$, such that $f \circ d_C = g \circ d_B$. Then, let $d_B^* : B \to D$ and $d_C^* : C \to D$ denotes the reversed ring homomorphism of the spaces, then one has $d_C' \circ f^* = d_b' \circ g^*$. Hence, going back to the top diagram, one has a unique ring homomorphism $h^* : B \otimes_A C \to D$ by the fibre coproduct property in CRing, hence generates a unique morphism of schems $h : W = \mathrm{Spec}(D) \to X \times_Z Y = \mathrm{Spec}(B \otimes_A C)$ (and any such morphism of schemes h must corresponds to h^*, showing the uniqueness):

$$\begin{tikzcd}
Z & X \arrow[l, "f"'] & \\
Y \arrow[u, "g"] & X \times_Z Y \arrow[l, "f'"] \arrow[u, "g'"'] & \\
& & W \arrow[ul, dotted, "\exists! h"'] \arrow[ull, bend left, "d_B"] \arrow[uul, bend right, "d_C"']
\end{tikzcd}$$

Basically, it's the reversed diagram of the very first one.

Finally, we ought to show its fibre product property in Sch. This can be done through affinization: Suppose W is arbitrary schemes, with $d_B : W \to Y$ and $d_C : W \to X$ satisfies $g \circ d_B = f \circ d_C$. The universal property of affinization guarantees a factorization $d_B' : \mathrm{Aff}_W \to Y$ and $d_C' : \mathrm{Aff}_W \to C$ as follow:

$$\begin{tikzcd}
Z & X \arrow[l, "f"'] & & \\
Y \arrow[u, "g"] & X \times_Z Y \arrow[l, "f'"] \arrow[u, "g'"'] & & \\
& & \mathrm{Aff}_W \arrow[ul, dotted, "\exists! h"'] \arrow[ull, bend left, "d_B'"] \arrow[uul, bend right, "d_C'"'] & \\
& & & W \arrow[ul, "a_W"'] \arrow[uulll, bend left, "d_B"] \arrow[uuull, bend right, "d_C"']
\end{tikzcd}$$

Where, the universality of $X \times_Z Y$ over affine schemes guarantees a unique morphism of schemes $h : \mathrm{Aff}_W \to X \times_Z Y$ for the diagram to commute, and composing with the affinization morphism $a_W : W \to \mathrm{Aff}_W$, it generates a morphism for W to satisfy the fibre product property. It is unique, simply because any morphism of schemes $W \to X \times_Z Y$ (where $X \times_Z Y$ is affine!) must also factor through Aff_W uniquely, and such factor must be unique by the fibre product in AffSch shown above. So, it's indeed a fibre product in Sch, not just in AffSch.

□

To show it works for general schemes, it's similar to before: Restrict them to affine neighborhoods, construct fibre product neighborhood-wise, then glue all of them together. For this, we need some machineries of gluging morphisms:

Lemma: Gluing of Morphisms